



A Multi-Objective Rice Supply Chain Model under Uncertainty: Minimizing Cost, Soil-Erosion Impact, and Water Consumption

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Abstract: Rice is a strategic food commodity, and its supply chain involves farms, mills, distribution centres, and markets. This study presented a multi-objective mathematical model for a rice supply chain that minimized total costs (economic objective), soil erosion caused by water used in cultivation (environmental objective), and total water consumption. A real case study from Iran including four major rice-producing regions was analysed under three water availability scenarios. To handle uncertainty in rainfall and irrigation water, stochastic programming was applied and the multi-objective model was solved using extended goal programming. Sensitivity analyses examined changes in water availability, objective function weights, and import limits. Results demonstrated that under water-scarce conditions, production in arid regions (e.g., Khuzestan) was not economically or environmentally viable, leading to increased imports. The model provides a practical tool for policymakers balancing food security, cost, environmental sustainability, and water conservation.

Keywords: Rice supply chain; Multi-objective; Environmental impact; Uncertainty; Stochastic programming; Extended goal programming

1 Introduction

Agriculture is a critical sector for a nation's economic, political, and social stability. However, the lack of integrated information systems and functional supply chain models has limited the effectiveness of this sector despite its potential [1, 2]. Agriculture contributes approximately 13% of food demand, 11% of raw materials for the food industry, 11% of gross national product, 22% of national employment, and 23% of non-oil exports in many developing economies [3]. Sustainable agriculture improves environmental quality and resource use while meeting human food needs [4].

Among agricultural products, rice holds a strategic position due to its high consumption and the environmental intensity of its cultivation. Proper design of the rice supply chain is essential for producing countries. This study proposed a multi-objective mathematical model for the rice supply chain that addressed both economic and environmental goals. The model sought to answer the following questions:

1. What is the optimal amount of unhusked rice production per region considering costs, demand, and environmental factors?
2. How much rice should be produced domestically versus imported?
3. What share of rice should go directly to markets or through distribution centres?
4. What is the amount of water consumption to reduce environmental impacts?
5. What is the optimal cultivated area for rice?

The proposed model was linear and multi-objective, thus minimizing total costs and environmental damage from water use. An extended goal programming approach was applied, while uncertainty in water availability was handled via stochastic programming. A case study from Iran, featuring four major rice-producing provinces and import options, validated the model.

2 Literature Review

Supply chain management integrates activities from raw material procurement to final product distribution [2, 5]. The agricultural supply chain has received growing attention due to its complexity, which includes cultivation, harvesting, processing, and distribution under uncertain conditions [6, 7].

Agriculture faces two key challenges: meeting growing demand and balancing production with environmental sustainability [8]. The main components of an agricultural or rice supply chain include: (1) production and storage; (2) processing in mills; (3) distribution to wholesalers or centres; and (4) delivery to customers [6].

Uncertainty is inherent in agricultural supply chains due to weather, soil conditions, water availability, and market fluctuations [9]. Operations research has addressed agricultural problems since the 1940s, but uncertainty modelling gained prominence only in the last two decades [10, 11]. Common uncertainty-handling techniques include stochastic programming, robust optimisation, and simulation [12].

Early work by Sorensen and Gilheany [13] simulated sugarcane harvest under stochastic weather conditions. Arnaout and Maatouk [14] minimized harvest and transportation costs for processing facility location. Gorton et al. [15] examined environmental factors in fresh produce supply chains, including government policies and resource impacts. Masud et al. [16] modeled water consumption and soil damage for barley cultivation in Canada under uncertain rainfall.

Sustainability in agricultural supply chains nowadays integrates economic, environmental, and social objectives. Reidsma et al. [17] reviewed development and use of farm models for policy assessment in the European Union. Stochastic programming is the most frequently used uncertainty method. Sazvar et al. [18] optimized a three-objective sustainable supply chain for organic products, to minimize costs and environmental damage while maximizing consumers' health.

Recent studies (2019–2025) have extended these themes. For example, Achmad et al. [19] developed an agent-based modelling and robust optimisation for food supply chains during the COVID-19 pandemic. McClements [20] developed a roadmap for future agriculture research designing a sustainable and environmentally friendly food supply chain. Mortazavizadeh et al. [21] applied machine learning to predict water demand in agricultural water management. Gonçalves et al. [22] proposed a blockchain-enabled traceability system for sustainable rice supply chains. Douaioui et al. [23] developed an innovative goal programming and advanced metaheuristic techniques in optimizing supply chain efficiency.

Several recent studies have explicitly incorporated water consumption as a key performance metric in agricultural supply chain design, namely Hajimirzajan et al. [24] fuzzy framework for water-land allocation, Accorsi et al. [25] ecosystem carbon and water balance, Siyal et al. [26] water footprint for supply chain irrigation, and Baghizadeh et al. [27] water-energy-food nexus.

To handle uncertainty in agricultural supply chains, Hosseini and Goli [28] proposed a robust optimisation model, while Alinezhad et al. [29] developed a fuzzy multi-objective optimisation. Hosseini-Motlagh et al. [30] presented a hybrid sustainability and resiliency in the wheat network.

Seyedzadeh et al. [31] designed a multi-objective sustainable network, while Daneshvar et al. [32] proposed a multi-objective integrated with meta-heuristic algorithms for agricultural sustainable supply chain. Reidsma et al. [17] reviewed farm models for policy impact assessment.

2.1 Research Gap and Innovations

According to the reviewed agricultural supply chains, most research focused on economic objectives (cost minimization or profit maximization). Environmental objectives appear to be less frequently and typically address pollution, resource use, or waste. Few studies specifically examined water consumption and soil erosion in rice supply chains under uncertainty.

2.2 Innovations of This Study

- Three objectives: minimizing total cost, water-related soil erosion, and total water consumption.
- Explicit modelling of surface water (rainfall) and irrigation water as separate sources.
- Rice-specific case study with four Iranian provinces.
- Extended goal programming combined with stochastic programming.
- Three water availability scenarios (bad, medium, and good).

3 Definition of the Problem

The proposed rice supply chain in Figure 1 consists of six levels: farms, rice mills, distribution centres, import centres, by-product markets, and final consumer markets. Unhusked rice is harvested at farms and transported to rice mills, where it is processed into white rice and by-products (rice bran and broken rice). White rice may be sent directly from rice mills to markets or routed through distribution centres for packaging and redistribution. Imported rice enters through import centres and moves to distribution centres, then to markets. Market demand is proportional

to regional population. Figure 1 illustrates the six-level supply chain structure: (1) farms (production sites); (2) rice mills (processing facilities); (3) distribution centres (storage and repackaging); (4) import centres (foreign supply sources); (5) by-product markets (rice bran and broken rice); and (6) final consumer markets. Arrows indicate product flows from farms to rice mills, from rice mills to distribution centres or markets, from import centres to distribution centres, and from distribution centres to final markets. The multi-objective model proposed by Kazemi et al. [33] serves as the foundation for this study. We extend their work by incorporating water consumption as a third objective, transforming the problem into a tri-objective formulation. The inclusion of water consumption as a separate objective is motivated by recent studies highlighting the critical role of water scarcity in agricultural supply chains [24, 26, 27]. As noted by Hajimirzajan et al. [24], integrated water-land allocation is essential for sustainable crop planning, and Baghizadeh et al. [27] emphasized the water-energy-food nexus in supply chain design.

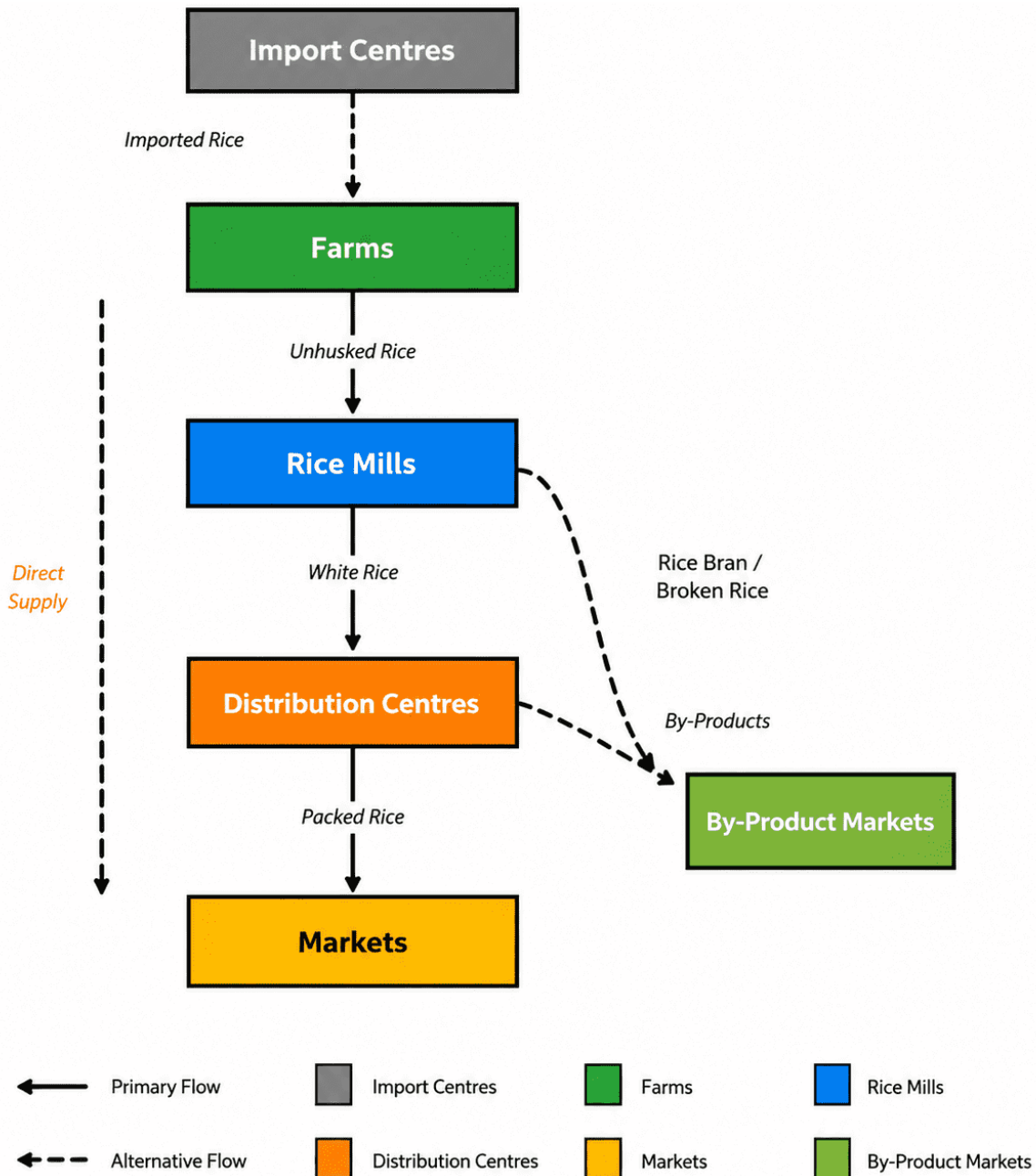


Figure 1. Schematic of the proposed rice supply chain

3.1 Mathematical Model

The model is linear with three objective functions: (1) minimize total costs; (2) minimize environmental damage from water use in cultivation; and (3) minimize water consumption. The indices are given in Table 1, parameters are

defined in Table 2, and decision variables are shown in Table 3.

Table 1. Indices and definitions

Index	Description
I	Farm $(1, \dots, I)$
J	Rice mill $(1, \dots, J)$
K	Distribution centre $(1, \dots, K)$
S	Import centre $(1, \dots, S)$
N	Market $(1, \dots, N)$
Q	By-product $(1, \dots, Q)$
C	Scenario $(1, 2, 3 \text{ for bad, medium, and good, respectively})$

Table 2. Parameters and definitions

Parameter	Description
c_{ij}	Transport cost: farm i to rice mill j
c_{jk}	Transport cost: rice mill j to distribution centre k
c_{jn}	Transport cost: rice mill j to market n
c_{sk}	Transport cost: import centre s to distribution centre k
c_{kn}	Transport cost: distribution centre k to market n
c_{jq}	Transportation cost of by-product q from rice mill j to by-product market
d_j	Processing capacity of rice mill j
v_k	Capacity of distribution centre k
λ_i	Production cost on farm i
p_j	Processing cost at rice mill j
ω_s	Import cost from centre s
dem_n	Rice demand in market n
dem'_q	Market demand for by-product q (tons)
cfe_i	Environmental damage coefficient for farm i (soil loss per m ³ water)
θ_i	Unhusked rice yield (ton/hectare) on farm i
su_i	Maximum cultivable area (hectares) on farm i
cwb_{ic}	Irrigation water coefficient under scenario c (m ³ /hectare) on farm i
cwg_{ic}	Rainfall water coefficient under scenario c (m ³ /hectare) on farm i
α_q	Conversion coefficient for by-product q (per ton of unhusked rice)
β	Rice processing coefficient (conversion rate from unhusked rice to white rice)
TW_i	Total available water at farm i over the planning horizon (m ³)
Wg_i	Water from rainfall on farm i (m ³) - Wg_{ic} stochastic form under scenario c
Wb_i	Water from irrigation on farm i (m ³) - Wb_{ic} stochastic form under scenario c

Table 3. Variables and definitions

Variable	Description
Sur_i	Cultivated area on farm i (hectares)
X_{ij}	Unhusked rice from farm i to rice mill j
Y_{jk}	Rice from rice mill j to distribution centre k
$Y_{jn'}$	Rice directly from rice mill j to market n
Z_{sk}	Imported rice from centre s to distribution centre k
Y_{jq}''	Quantity of by-product q produced at rice mill j (tons)
W_{kn}	Rice from distribution centre k to market n
Cwr_{ic}	Total water used on farm i under scenario c

3.1.1 Objective functions

The three objective functions of the proposed model are formulated as follows:

• **Minimize Total Costs:**

$$\begin{aligned} \min \text{ Costs} = & \sum_i \sum_j X_{ij} c_{ij} + \sum_j \sum_k Y_{jk} c_{jk} + \sum_j \sum_n Y'_{jn} c_{jn} + \sum_j \sum_q Y''_{jq} c_{jq} + \sum_s \sum_k Z_{sk} c_{sk} + \\ & \sum_k \sum_n W_{kn} c_{kn} + \sum_i \sum_j X_{ij} \lambda_i + \sum_j \sum_k Y_{jk} p_j + \sum_j \sum_n Y'_{jn} p_j + \sum_s \sum_k Z_{sk} \omega_s \end{aligned} \quad (1)$$

• **Minimize Environmental Impact (Soil Erosion):**

$$\min \text{ Env} = \sum_i (cfe_i \times Cwr_i)$$

Under uncertainty, the environmental objective becomes:

$$\min \text{ Env} = \sum_c \sum_i p_c \times (cfe_i \times Cwr_{ic}) \quad (2)$$

where, p_c is the probability of scenario c .

• **Minimize Total Water Consumption:**

$$\min Z_3 = \sum_i Cwr_i = \sum_i (Wg_i + Wb_i)$$

where, water consumption is already calculated in the model via $Cwr_i = Wg_i + Wb_i$.

Under uncertainty, the water consumption objective becomes:

$$\min Z_3 = \sum_c \sum_i p_c \times Cwr_{ic} \quad (3)$$

3.1.2 Constraints

The constraints of the proposed model are given as follows:

$$\beta \sum_i X_{ij} = \sum_k Y_{jk} + \sum_n Y'_{jn}, \quad \forall j \quad (4)$$

$$\sum_j Y_{jk} + \sum_s Z_{sk} = \sum_n W_{kn}, \quad \forall k \quad (5)$$

$$\sum_j X_{ij} \leq \theta_i \times Sur_i, \quad \forall i \quad (6)$$

$$\sum_i X_{ij} \leq d_j, \quad \forall j \quad (7)$$

$$\sum_j Y_{jk} + \sum_s Z_{sk} \leq v_k, \quad \forall k \quad (8)$$

$$Sur_i \leq su_i, \quad \forall i \quad (9)$$

$$\sum_q \alpha_q + \beta = 1 \quad (10)$$

$$\sum_i X_{ij} \cdot \alpha_q = Y''_{jq} \quad \forall j, q \quad (11)$$

$$\sum_j Y''_{jq} \geq dem'_q \quad \forall q \quad (12)$$

Eqs. (4)–(9) define the rice mill balance (no waste), distribution centre balance, farm production limit, rice mill capacity, distribution centre capacity and cultivated area limit constraints, respectively. All decision variables are non-negative. Eqs. (10)–(12) indicate by-product related constraints, Eq. (10) implies conversion balance, Eq. (11) confines by-product generation, and Eq. (12) ensures by-product demand satisfaction.

4 Approach to the Solution

The proposed model involved two complexities: (1) uncertainty in water availability (rainfall and irrigation); and (2) three conflicting objectives (cost vs. environmental impact and water consumption). We addressed uncertainty with stochastic programming and the multi-objective nature with extended goal programming. Multi-objective optimisation under uncertainty has been extensively studied in recent agricultural supply chain literature. Hosseini and Goli [28] employed a robust optimisation approach, while Alinezhad et al. [29] used fuzzy multi-objective programming. The extended goal programming method adopted in this study is consistent with these recent advances and has been successfully applied in similar contexts [17].

4.1 Stochastic Programming

In real-world agricultural problems, parameters such as rainfall and groundwater availability are uncertain. Stochastic programming provides a framework for optimizing under uncertainty by defining a set of scenarios, each with a probability of occurrence.

Let $\Omega = \{\omega_1, \omega_2, \dots, \omega_s\}$ represent the set of scenarios. Each scenario ω_k occurs with probability $p(\omega_k)$, where:

$$p(\omega_k) \geq 0, \quad \sum_{k=1}^s p(\omega_k) = 1 \quad (13)$$

In this study, three scenarios were defined for water availability:

Good scenario: high rainfall and low irrigation need.

Medium scenario: moderate rainfall and irrigation.

Bad scenario: low rainfall and high irrigation need.

The deterministic equivalent of the stochastic model replaced the original environmental objective with the expected value across scenarios:

$$\min \text{Env} = \sum_c \sum_i p_c \times (cfe_i \times Cwr_{ic}) \quad (14)$$

Water-related constraints become scenario-dependent:

$$\begin{aligned} Cwr_{ic} &= Wg_{ic} + Wb_{ic} & \forall i, c \\ Wg_{ic} &= cw_{gic} \times Sur_i & \forall i, c \\ Wb_{ic} &= cw_{bic} \times Sur_i & \forall i, c \\ Cwr_{ic} &\leq TW_i & \forall i, c \end{aligned} \quad (15)$$

All other constraints remain deterministic.

Three scenarios are defined for water availability: Bad, Medium, and Good. The Stochastic solution is not a separate scenario but rather the expected value solution obtained by solving the stochastic programming model with equal probabilities (1/3 each) assigned to the three scenarios. This is included in the results for comparison with the scenario-specific solutions.

4.2 Extended Goal Programming

To solve the multi-objective model, we used the extended goal programming method proposed by Romero [34]. This method balances the achievement of multiple goals and allows the decision-maker to control the trade-off between optimisation and goal satisfaction through a parameter α .

The general formulation is:

$$\begin{aligned} \min \quad & \alpha\lambda + (1 - \alpha) \sum_{i=1}^q \left(\frac{u_i n_i}{b_i} + \frac{v_i p_i}{b_i} \right) \\ \text{s.t.} \quad & \frac{u_i n_i}{b_i} + \frac{v_i p_i}{b_i} \leq \lambda, \quad i = 1, \dots, q \\ & f_i(x) + n_i - p_i = b_i, \quad i = 1, \dots, q \\ & x \in F, \quad n_i, p_i \geq 0 \end{aligned} \quad (16)$$

where,

q = number of objectives (here $q = 3$);

b_i = target value for objective i (obtained by solving each objective separately);

n_i, p_i = negative and positive deviations from the target;

u_i, v_i = weights for deviations;

λ = maximum weighted deviation;

$\alpha \in [0, 1]$ = trade-off parameter ($\alpha = 1$ emphasizes goal achievement; $\alpha = 0$ emphasizes optimisation).

In this study, the extended goal programming model is formulated as follows:

$$\begin{aligned}
\min \quad & \alpha\lambda + (1 - \alpha) \left[\left(\frac{u_1 n_1}{b_1} + \frac{v_1 p_1}{b_1} \right) + \left(\frac{u_2 n_2}{b_2} + \frac{v_2 p_2}{b_2} \right) + \left(\frac{u_3 n_3}{b_3} + \frac{v_3 p_3}{b_3} \right) \right] \\
\text{s.t.} \quad & \frac{u_1 n_1}{b_1} + \frac{v_1 p_1}{b_1} \leq \lambda \\
& \frac{u_2 n_2}{b_2} + \frac{v_2 p_2}{b_2} \leq \lambda \\
& \frac{u_3 n_3}{b_3} + \frac{v_3 p_3}{b_3} \leq \lambda \\
& f_1(x) + n_1 - p_1 = b_1 \\
& f_2(x) + n_2 - p_2 = b_2 \\
& f_3(x) + n_3 - p_3 = b_3 \\
& n_1, p_1, n_2, p_2, n_3, p_3 \geq 0
\end{aligned}$$

subject to the model constraints.

The weights u_1, v_1 for the economic objective, u_2, v_2 for the environmental objective and u_3, v_3 for the water consumption objective reflect policy priorities. The weights (u_1, v_1) represent the relative importance of achieving each objective's target value in the goal programming formulation. In the base case, we set $(u_1, v_1) = (0.6, 0.2)$ for the economic objective, $(u_2, v_2) = (0.2, 0.4)$ for the environmental objective, and $(u_3, v_3) = (0.2, 0.4)$ for the water consumption objective. These weights are used to generate the base-case goal programming solution reported in the results section. For comparison, the Min. cost only, Min. env only, and Min. water only solutions are obtained by solving single-objective problems with full weight assigned to each respective objective. The sensitivity analysis later varies these weights and the import-limit settings to examine their impacts on the solution.

5 Case Study

5.1 Case Description

The model was validated using data from Iran, where 17 out of 31 provinces produce rice, with annual white rice production of approximately 2.5 million tons. Four major producing regions were considered: Mazandaran, Gilan, Khuzestan, and Golestan. The supply chain included 4 farms (one per region), 4 rice mills (one per farm location), 6 distribution centres, 4 import centres, 32 market regions (provinces), and 2 by-products (rice bran and broken rice). Water scenarios are given in Table 4.

Table 4. Water scenarios

Scenario	Rainfall	Irrigation Use
Bad	Low	High
Medium	Moderate	Moderate
Good	High	Low

Key Assumptions. The main assumptions of the case study are as follows:

- One rice mill per farm; inter-region transfers allowed.
- Transport vehicles are identical; costs are known and fixed.
- Processing coefficients for white rice and by-products are constant across regions.
- Import volume is limited to a percentage of total demand.
- Water consumption is measured in m³ per hectare.
- Environmental damage is calculated based on surface water (rainfall + irrigation).
- Scenario probabilities are assumed to be equal, with each scenario assigned a probability of 1/3 unless otherwise stated.

Additional model parameters. Further key parameters of the model were extracted from the data as shown in Table 5.

The rice mill processing cost ranges from 300 to 315 thousand Tomans per ton, the import cost ranges from 1,080 to 2,280 thousand Tomans per ton, distribution centre capacities range from 190,000 to 350,000 tons, and annual rice demand is calculated as 36 kg per capita multiplied by the provincial population.

Table 5. Key parameters

Region	Max Cultivation (ha)	Yield (ton/ha)	Production Cost (1,000 Tomans/ton)	Environmental Damage Factor
Mazandaran	291,666	4.8	125	0.004
Gilan	234,000	5.0	115	0.006
Khuzestan	100,000	4.0	130	0.05
Golestan	163,000	4.3	120	0.03

5.2 Results and Analysis

The multi-objective model was coded in Linear Interactive and General Optimizer (LINGO) optimisation software and solved using extended goal programming with the following weights. In the base case, the extended goal programming weights are set as $(u_1, v_1) = (0.6, 0.2)$ for the economic objective, $(u_2, v_2) = (0.2, 0.4)$ for the environmental objective, and $(u_3, v_3) = (0.2, 0.4)$ for the water consumption objective. These weights reflect the relative importance of minimizing cost, soil erosion, and water usage, respectively. Sensitivity analyses in Section 5.3 vary these weights to examine their impact on the solution. Table 6 compares the results of solving the multi-objective model using four different approaches: (1) minimizing only total costs (economic objective); (2) minimizing only environmental impact; (3) minimizing only water consumption; and (4) extended goal programming balancing all objectives. The table reports cost objective (Z1), environmental impact objective (Z2), and water objective (Z3), production percentage, import percentage, and deviations from individual optima.

The goal programming solution balances three objectives, with each deviating as reported in Table 6 from its individual optimum. This represents a practical trade-off for policymakers.

Table 6. Objective function values under different solution modes

Solution Mode	Z1 (Cost)	Z2 (Env.)	Z3 (Water)	Production (%)	Import (%)	Deviations from Z1	Deviations from Z2	Deviations from Z3
Min. cost only	5.02B	10.2	185.6M	74%	26%	0%	145%	132%
Min. env only	16.18B	4.88	142.3M	61%	39%	222%	0%	78%
Min. water only	15.85B	5.12	128.7M	58%	42%	216%	5%	0%
Goal programming	5.68B	4.92	152.4M	66.5%	33.5%	13%	1%	18%

Note: Z1: cost objective; Z2: environmental impact objective; Z3: water objective. Units: Z1 in 1,000 Tomans; Z2 in dimensionless soil loss units; Z3 in m³. B: billion; M: million. Env.: environmental impact.

Figure 2 shows the optimal production quantities for the four major rice-producing regions in Iran: Mazandaran (1,400,000 tons), Gilan (1,170,000 tons), Khuzestan (0 ton), and Golestan (approximately 650,000 tons). The figure demonstrates that Khuzestan produces no rice due to its high environmental damage coefficient (0.05) and water scarcity, which is consistent with regional water-scarcity restrictions.

Figure 3 illustrates the distribution of rice shipments from each rice mill to the six distribution centres (DC). The figure shows that distribution centres 4, 5 and 6 receive 0 shipment from domestic mills, indicating that these centres are served primarily by import channels.

Table 7 shows the distribution of imported rice from four import centres to six distribution centres. The data indicate the linkage between import centres and their respective serving distribution centres as well as the corresponding shipment volumes. For example, Import Centre 1 supplies DC3 (350,000 tons), DC4 (60,000 tons), and DC5 (190,000 tons). DCs stand for distribution centres.

Markets receive rice via two channels: Directly from rice mills or through distribution centres. Major producing provinces such as Mazandaran and Gilan had higher direct purchase coefficients (0.6–0.9), reflecting local supply chains. Table 8 presents sample direct supply quantities from rice mills to specific markets. The two major producing provinces had higher direct purchase coefficients (0.6–0.9), reflecting shorter supply chains and local distribution patterns. For example, Mazandaran mill supplied 118,080 tons to Mazandaran market, and 336,095 tons to Tehran market.

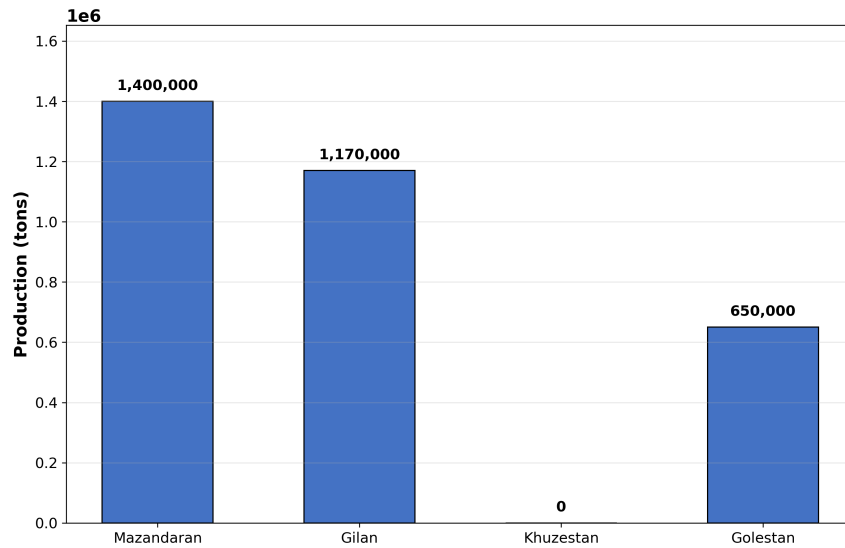


Figure 2. Amount of unhusked rice produced

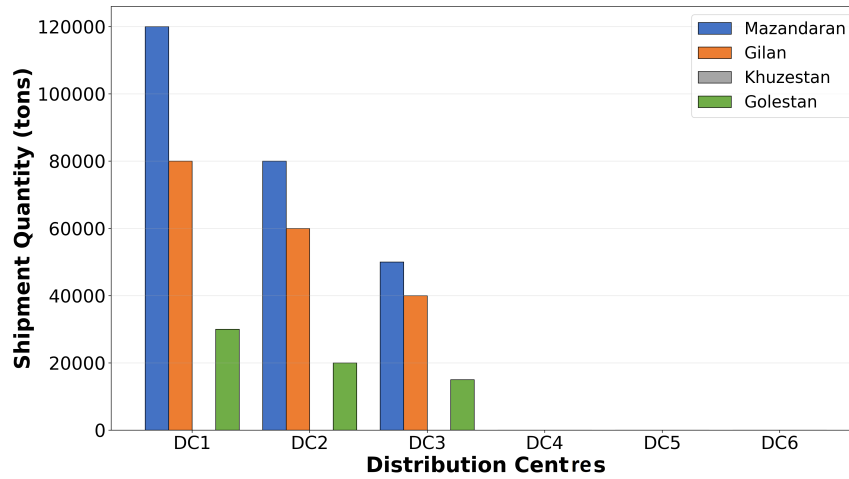


Figure 3. Amount of rice transferred from mills to distribution centres (DC)

Table 7. Imported rice to distribution centres (DC) (tons)

Import Centre	DC1	DC2	DC3	DC4	DC5	DC6
1	0	0	350,000	60,000	190,000	0
2	17,334	76,309	0	0	0	20,000
3	0	0	0	0	100,000	0
4	100,000	0	0	0	0	0

Table 8. Direct supply of rice from rice mills to selected markets

Market	Rice Mill Source	Amount (tons)
Mazandaran	Mazandaran mill	118,080
Gilan	Gilan mill	54,000
Tehran	Mazandaran mill	336,095

Table 9 quantifies water use and environmental impact for the four rice-producing regions. It shows rainfall water (m^3), irrigation water (m^3), total water consumption (m^3), and soil erosion (m^2 soil loss) calculated as total

water multiplied by the environmental damage coefficient ($cf e_i$). Khuzestan showed zero production under the base solution due to its high damage coefficient (0.05 vs. 0.006 for northern regions).

Table 9. Water use and soil erosion by region (medium scenario)

Region	Rainfall Water (m ³)	Irrigation Water (m ³)	Total Water (m ³)	Soil Erosion (m ² soil loss)
Mazandaran	140,000,000	76,000,000	216,000,000	864,000
Gilan	120,000,000	65,000,000	185,000,000	1,110,000
Khuzestan	0	0	0	0
Golestan	70,000,000	40,000,000	110,000,000	3,300,000

Note: Soil erosion = total water $\times$ $cf e_i$ (environmental damage coefficient).

5.3 Sensitivity Analyses

Analysis 1 (Water Scenarios)

Figure 4 shows the quantities of rice bran and broken rice produced at each rice mill. These by-products were generated during the milling process and were transferred and sold to separate markets, thus contributing to overall supply chain costs.

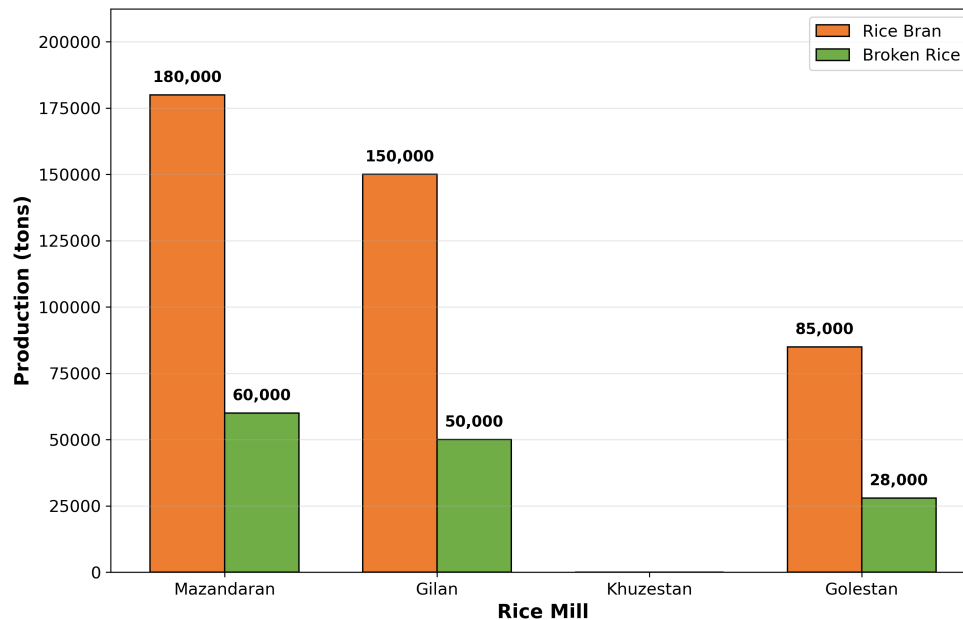


Figure 4. Comparison of by-products produced in different provinces

Table 10 compares three water scenarios (bad, medium, and good) across multiple performance metrics. The results showed that Khuzestan produced rice only under the good water scenario. As water conditions improved, domestic production increased from 61% to 71%, imports decreased from 39% to 29%, and environmental impact (Z2) increased (worsened) due to higher soil erosion in raising and harvesting more rice. Z3 decreases as water conditions improve because higher rainfall reduces irrigation requirements.

Table 10. Sensitivity analysis—water scenarios

Scenario	Production (%)	Import (%)	Z1	Z2	Z3	Khuzestan Production?
Bad	61%	39%	5.92B	4.05	165.2M	No
Medium	66.5%	33.5%	5.68B	4.92	152.4M	No
Good	71%	29%	5.55B	5.68	138.7M	Yes (215,000 tons)
Stochastic	66%	34%	5.70B	4.88	148.9M	No

Note: Z1: cost objective; Z2: environmental impact objective; Z3: water objective. Units: Z1 in 1,000 Tomans; Z2 in dimensionless soil loss units; Z3 in m³. B: billion; M: million.

Only under the good water scenario did Khuzestan produce rice. Production increased and import decreased as water conditions improved. The Stochastic solution in Table 10 represents the expected value solution obtained by solving the stochastic programming model with equal probabilities (1/3 each) assigned to the Bad, Medium, and Good scenarios. For this dataset, the expected solution falls near the scenario-specific medium case, because the equal-probability weighting of the three scenarios yields expected values that match the medium scenario parameters. This is a characteristic of the specific input data and does not indicate an error.

Analysis 2: Importance of Objective Functions (weights u, v)

Figure 5 compares provincial production quantities under different water scenarios. The results show that Khuzestan produces rice only under the good water scenario, while production in the northern provinces remains stable.

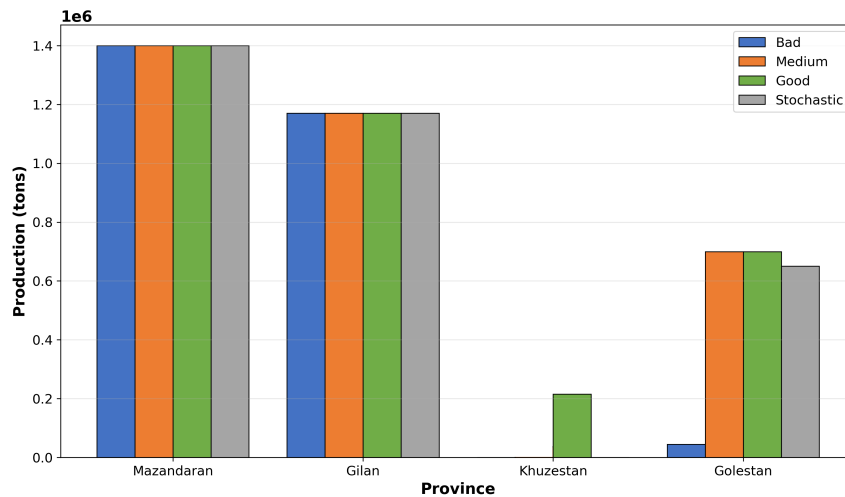


Figure 5. Production in different scenarios for provinces

Table 11 shows how production percentage, import percentage, and deviations from individual optima changed as the relative importance of the economic, environmental, and water consumption.

Table 11. Sensitivity analysis—objective function weights

Economic Weight	Environmental Weight	Water Weight	Production (%)	Import (%)	Z1 Deviations	Z2 Deviations	Z3 Deviations
0.8	0.1	0.1	70%	30%	6%	72%	65%
0.6	0.2	0.2	68%	32%	11%	45%	42%
0.5	0.3	0.2	66.5%	33.5%	15%	15%	18%
0.4	0.3	0.3	63%	37%	20%	8%	10%
0.3	0.3	0.4	61%	39%	22%	5%	5%
0.2	0.3	0.5	60%	40%	24%	4%	3%

Note: Z1: cost objective; Z2: environmental impact objective; Z3: water objective.

As environmental and water importance increased, production decreased (import increased) to reduce water-related soil erosion.

Analysis 3: Import Limit

Figure 6 presents how production and import shares (%) under Bad, Medium, Good, and Stochastic water scenarios, showing that production increases from 61% under the bad scenario to 71% under the good scenario, while imports decrease from 39% to 29%.

Table 12 demonstrates the effect of reducing the maximum allowable import limit on production ratio, import ratio, Khuzestan production status, and deviations from individual optima.

Reducing the import limit forces domestic production to increase, including production from Khuzestan, and increases environmental and water-related deviations. The model becomes infeasible only when the maximum import rate is reduced below 24%.

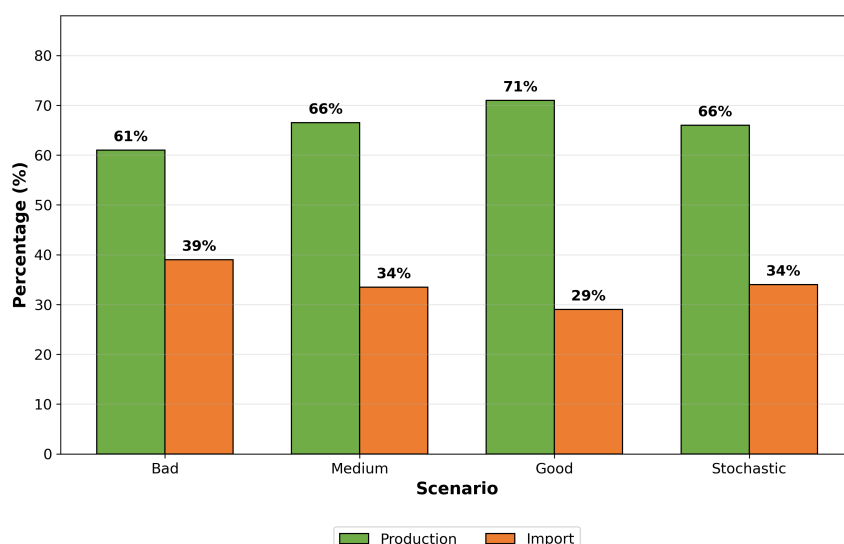


Figure 6. Production and import shares for different scenarios

Table 12. Results of sensitivity analysis for varying limits of maximum import rate

Max Import Rate	Production (%)	Import (%)	Z1 Deviations	Z2 Deviations	Z3 Deviations	Khuzestan Production?
50% and more	66.5%	33.5%	13%	1%	18%	No
40%	66.5%	33.5%	13%	1%	18%	No
35%	66.5%	33.5%	13%	1%	18%	No
30%	70%	30%	11%	45%	42%	Yes (48,000 tons)
27%	73%	27%	6%	86%	78%	Yes (215,000 tons)
25%	75%	25%	2.6%	116%	95%	Yes (324,555 tons)
24%	76%	24%	0.9%	131%	115%	Yes (372,482 tons)
< 24%	—	—	—	—	—	Infeasible

Note: Z1: cost objective; Z2: environmental impact objective; Z3: water objective. Units: Z1 in 1,000 Tomans; Z2 in dimensionless soil loss units; Z3 in m³. B: billion; M: million. “—” indicates no data.

6 Conclusions

Rice is a strategic crop providing food for over half of the world’s population. For producing countries, an efficient rice supply chain supports both food security and economic growth. This study presented a multi-objective mathematical model for a rice supply chain that minimized total costs (economic) and soil erosion caused by water used in cultivation (environmental) and total water consumption—this study contributes to rice supply chain modelling by explicitly including water minimization as a separate objective. Uncertainty in rainfall and irrigation water was handled through stochastic programming with three scenarios (bad, medium, and good); the multi-objective model was solved using extended goal programming.

A case study from Iran including 4 major rice-producing provinces validated the model. Under base conditions as stated in Sections 4.2 and 5.2, the goal programming solution achieved a balanced trade-off with 13% deviation from the cost optimum, 1% from the environmental optimum, and 18% from the water optimum. Khuzestan province did not produce rice in the base solution due to its high environmental damage coefficient (0.05) and high water consumption, consistent with regional water-scarcity restrictions. Under the “good” water scenario, production in Khuzestan becomes viable (215,000 tons) with reduced water consumption (138.7M m³) compared to the bad scenario (165.2M m³), demonstrating that improved rainfall conditions lower irrigation requirements. The water minimization objective (Z3) significantly influences production decisions: when water weight increases to 0.5, production decreases to 60% and imports increase to 40%. Reducing the maximum import limit below 24% renders the model infeasible, revealing a practical constraint on import substitution. The model achieves significant water savings: compared to the cost-only solution, the balanced solution reduces water consumption by 17.9% (from 185.6M m³ to 152.4M m³) with only a 13% increase in cost.

6.1 Future Research Directions

1. Incorporate additional objectives such as profit maximization, resource utilization, or social welfare;
2. Add an export sector if domestic production exceeds demand;
3. Model demand uncertainty (e.g., population growth and consumption shifts) alongside supply uncertainty; and
4. Apply robust optimisation or fuzzy programming to further address parameter uncertainty.

Data Availability

The data supporting our research results are included within the article.

Conflicts of Interest

The author declares no conflicts of interest.

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